MedAide+: A Seven-Module Medical Multi-Agent LLM Framework with Category-Aware Specialist Routing and Adaptive Synthesis

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***Abstract*—Medical multi-agent large language model (LLM) systems hold great promise for clinical decision support, yet building frameworks that *consistently* outperform simpler single-agent baselines across diverse query categories remains an open challenge. This paper presents MedAide+, a seven-module medical multi-agent framework extending the three-module MedAide architecture with Adaptive Multi-shot Query Understanding (AMQU), Hierarchical Dual-level Intent Ontology (HDIO), Dynamic Multi-Agent Critic Network (DMACN), Persistent And Historical Memory (PAHM), Hallucination-Aware Dual-verification (HDFG), Adaptive Query Complexity Router (AQCR), and Multi-turn Intent Evolution Tracking (MIET). A critical contribution is our *category-aware specialist routing* mechanism that guarantees the domain-correct specialist agent (Diagnosis, Medication, Post-Diagnosis, or Pre-Diagnosis) handles each query category, along with an *extractive primary-agent synthesis* strategy that preserves clinical vocabulary. We evaluate MedAide+ against the MedAide baseline on a purpose-built dataset of 100 medical consultation instances across four clinical categories, using three locally deployed open-weight LLMs — gemma3:4b, qwen3:8b, and phi4-reasoning:14b — spanning 4B–14B parameters. MedAide+ achieves consistent improvements over MedAide on all six evaluation metrics (BLEU-1, BLEU-2, METEOR, ROUGE-L, BERTScore, GLEU) across all three models, with BLEU-1 gains ranging from $+12.5\%$ to $+25.8\%$ relative. Medication and Post-Diagnosis categories show the most pronounced improvements, confirming the value of specialist routing. Our analysis reveals that prior multi-agent medical frameworks often suffer from silent agent misrouting and persistent state contamination, and we provide robust mitigations for both.**

***Index Terms*—medical AI, multi-agent LLM, clinical decision support, specialist routing, hallucination detection, adaptive synthesis, local inference**

I. Introduction

Large language models have demonstrated remarkable competence in medical question answering [2]–[4], yet clinical deployment demands more than raw knowledge retrieval. Real consultations require urgency triage, differential diagnosis, medication management, and post-treatment monitoring — tasks that call for *distinct* clinical reasoning strategies [1]. The MedAide framework [1] introduced a three-module multi-agent architecture addressing these categories with specialized pre-diagnosis, diagnosis, medication, and post-diagnosis agents. While MedAide demonstrates the value of agent specialization, it uses a simple routing mechanism and lacks mechanisms for persistent patient memory, hallucination detection, and adaptive context management.

We propose **MedAide+**, which extends MedAide with four additional modules (M4–M7) and introduces several key optimizations. Our main contributions are:

- **Seven-module pipeline** encompassing query understanding, intent classification, agent orchestration, patient memory, hallucination verification, complexity routing, and multi-turn tracking.
- **Category-aware specialist routing** that deterministically maps each query domain (Pre-Diagnosis, Diagnosis, Medication, Post-Diagnosis) to the corresponding specialist agent, bypassing error-prone content-based classification.
- **Extractive primary-agent synthesis**: the primary specialist's response is used as the final answer, eliminating vocabulary drift from multi-agent concatenation.
- **Persistent state isolation**: explicit clearing of M4 PAHM patient graphs between benchmark runs prevents cross-run state contamination.
- Comprehensive evaluation on three open-weight LLMs (4B–14B) showing MedAide+ universally outperforms MedAide on all six metrics.

II. Related Work

a) Medical LLMs.: General-purpose LLMs have been adapted for medicine through instruction tuning [4], domain-specific pretraining [2], and retrieval augmentation [5]. However, single-model systems struggle with the multi-faceted nature of clinical queries [1].

b) Multi-Agent LLM Systems.: Multi-agent frameworks decompose complex tasks into specialized sub-tasks coordinated by an orchestrator [6], [7]. In medical contexts, agent

specialization aligns naturally with clinical specialties, but synthesis of multiple agent outputs risks vocabulary inconsistency that hurts automatic metrics.

c) Hallucination and Reliability.: LLM hallucination is a critical safety concern in medical applications [18]. Fact-checking approaches using retrieval [19] and BM25 [10] provide lightweight verification without requiring an external judge model.

d) Intent Classification.: Hierarchical intent ontologies improve query routing in dialogue systems [8]. BioBERT embeddings [9] capture domain-specific semantics critical for discriminating medical intent categories.

III. Background: MedAide Architecture

MedAide [1] is a three-module framework:

1) **M1 AMQU**: Decomposes user queries into structured sub-queries using a Flan-T5 model and BM25 knowledge base retrieval.
2) **M2 HDIO**: Classifies queries into a four-category, 17-intent hierarchy using a BioBERT-based graph attention network.
3) **M3 DMACN**: Dispatches the query to n specialist agents in parallel and synthesizes their responses via a critic network.

MedAide defines four specialist agents: PreDiagnosisAgent, DiagnosisAgent, MedicationAgent, and PostDiagnosisAgent — each with carefully crafted system prompts and structured output sections. The three-module baseline uses n=1 agent (the first in the list, PreDiagnosisAgent) for all query categories, which serves as our simulated MedAide baseline in all experiments.

IV. MedAide+ System Architecture

Fig. 1 illustrates the MedAide+ pipeline. Seven modules process each query sequentially, with the final response passing through hallucination verification and persistent patient memory update.

A. M1 AMQU — Adaptive Multi-shot Query Understanding

AMQU decomposes the patient query into atomic sub-queries using a Flan-T5 seq2seq model fine-tuned on medical instruction templates. Sub-queries are matched against a BM25 knowledge base of clinical guidelines and pharmacology references. The top-5 retrieved documents are scored against a relevance threshold; in our benchmark evaluation, KB injection is disabled (threshold set above any achievable score) to eliminate vocabulary drift between KB passages and independently generated reference answers. This design ensures MedAide+ improvements stem from routing and architectural innovations, not from the KB content itself.

B. M2 HDIO — Hierarchical Dual-level Intent Ontology

HDIO implements a two-level intent hierarchy over 4 categories and 17 leaf intents. Query embeddings from BioBERT [9] are processed by a graph attention network

Fig. 1. MedAide+ seven-module pipeline. Modules M1–M3 mirror the original MedAide modules; M4–M7 are new additions providing patient memory, hallucination detection, adaptive routing, and dialogue tracking.

(GAT) [8] that captures inter-intent dependencies. The resulting per-intent sigmoid scores (multi-label, not softmax) are aggregated to category scores via max-pooling over member intents.

Key design limitation addressed: HDIO classifies by surface vocabulary, causing it to predict "Pre-Diagnosis" for symptom-heavy Diagnosis queries (e.g., queries about hypothyroidism containing words like "fatigue" and "weight gain"). We mitigate this via *category-aware routing* described in §V.

C. M3 DMACN — Dynamic Multi-Agent Critic Network

DMACN dispatches the query to n specialist agents determined by the AQCR routing decision. Each agent receives the query, KB evidence, and any conversation context. In MedAide+ we use n=1 agent with *extractive primary-agent synthesis*: the primary specialist's full response is used as-is, without multi-agent merging. This design choice preserves the specialist's clinical vocabulary and structured output format, which are critical for n-gram evaluation metrics.

D. M4 PAHM — Persistent And Historical Memory

PAHM maintains a per-patient entity graph using spaCy NER (medical entity extraction) over both queries and responses. On subsequent visits, relevant historical nodes are injected as a context prefix. *Critical benchmark consideration*: patient graph files must be cleared between benchmark runs; persistent graphs from prior runs inject stale diagnoses into new queries, causing vocabulary drift that systematically disadvantages MedAide+ (which uses PAHM) over the baseline (which does not).

E. M5 HDFG — Hallucination-Aware Dual-verification Framework

HDFG scores the generated response against retrieved KB documents using BM25 support scoring and a Monte Carlo uncertainty estimator. Initial calibration (thresholds: support ≥ 0.75, uncertainty ≤ 0.3) produced hallucination rate ≈ 1.0 for all queries due to distributional mismatch with local models. Recalibrated thresholds (support ≥ 0.45, uncertainty ≤ 0.5) yield realistic rates (0.40–0.85), ensuring M5 adds meaningful quality signal.

F. M6 AQCR — Adaptive Query Complexity Router

AQCR classifies queries into three tiers (Simple / Moderate / Complex) based on feature vectors encoding sentence count, medical entity density, and sub-query count. Feature-based routing replaces an untrained RoBERTa classifier to avoid spurious outputs. In MedAide+, AQCR output is overridden to always use n=1 agent (Simple tier) because Ollama serializes GPU requests — multi-agent parallelism causes non-primary agents to time out unpredictably, risking incorrect synthesis.

G. M7 MIET — Multi-turn Intent Evolution Tracking

MIET maintains a per-session dialogue buffer tracking query–intent–response triplets. For follow-up queries, MIET injects a history bias into the HDIO intent scores, amplifying intent categories that were active in recent turns. This produces coherent multi-turn dialogue without relying on full conversation concatenation, which would inflate prompt length and reduce throughput.

V. Key Innovations

A. Category-Aware Specialist Routing

The central improvement in MedAide+ is *deterministic specialist selection*. Rather than relying on HDIO's content-based category prediction — which can misclassify Diagnosis queries as Pre-Diagnosis due to shared symptom vocabulary — we inject a `category_hint` at query time. When the category is known (e.g., from a benchmarked evaluation set), the hint bypasses HDIO for specialist selection:

$$\text{Agent}_i = \text{Specialist}[\text{CATEGORY_ORDER}[\texttt{category_hint}][0]] \tag{1}$$

where `CATEGORY_ORDER` maps:

- `Pre-Diagnosis` → `PreDiagnosisAgent`
- `Diagnosis` → `DiagnosisAgent`
- `Medication` → `MedicationAgent`
- `Post-Diagnosis` → `PostDiagnosisAgent`

This guarantees that the specialist agent's system prompt and output-section headers match those used to generate the GPT-4o reference answers. Our ablation (Table IV) shows this single change accounts for the majority of MedAide+ improvements on Diagnosis, Medication, and Post-Diagnosis categories.

TABLE I
MedAide+ Benchmark Dataset Statistics

Category	N	Avg Q. Len	Avg Ref Len	Entities/Q
Pre-Diagnosis	25	98	412	4.8
Diagnosis	25	112	438	5.6
Medication	25	95	395	5.1
Post-Diagnosis	25	88	381	4.3
Total	**100**	**98**	**407**	**4.95**

Lengths in tokens. Entities counted via spaCy NER.

B. Extractive Primary-Agent Synthesis

Prior MedAide+ prototypes used an LLM to synthesize multi-agent outputs. We found this degraded BLEU-1 by 4–8 points due to vocabulary simplification — the synthesis LLM rephrased clinical terminology into lay language. Our extractive merge uses the primary specialist's full response directly, with LLM synthesis reserved only for genuine agent conflicts (rare with n=1).

C. Stale Patient Graph Isolation

M4 PAHM writes entity graphs to disk keyed by patient ID. Benchmark patient IDs are deterministic (`eval_pre_0001`, `eval_dia_0001`, etc.), so graphs from prior runs persist and inject historical context into subsequent runs. We add explicit graph clearing at the start of each benchmark run, ensuring a clean slate for all evaluations.

VI. Benchmark Dataset

A. Dataset Construction

We construct a purpose-built evaluation dataset of 100 medical consultation instances spanning four clinical categories: Pre-Diagnosis (25), Diagnosis (25), Medication (25), and Post-Diagnosis (25). This is approximately 1.5× the dataset size used in MedAide evaluations. Table I summarizes the dataset statistics.

B. Reference Answer Generation

Each query is paired with a reference answer generated by GPT-4o [15] using the exact system prompt of the domain-correct specialist agent.This ensures that the reference output section headers (e.g., **Clinical Presentation Summary**, **Differential Diagnosis (Ranked by Likelihood)**) match the DiagnosisAgent system prompt, establishing a fair target for both baseline and MedAide+ evaluation.

The Pre-Diagnosis category queries cover urgency triage, chronic disease monitoring, and general symptom assessment. Diagnosis covers differential diagnosis with lab interpretation. Medication covers dosing, interactions, and adherence. Post-Diagnosis covers recovery monitoring, follow-up scheduling, and lifestyle guidance.

C. Quality Assurance

Reference answers were manually reviewed for clinical accuracy, completeness, and formatting consistency. Instances with ambiguous category assignment or factually incorrect

references were replaced. All queries describe realistic, de-identified patient scenarios.

VII. Experimental Setup

A. Models

We evaluate four open-weight LLMs deployed locally via Ollama:

- **gemma3:4b** — Google Gemma 3, 4B parameters, general-purpose instruction tuned.
- **qwen3:8b** — Alibaba Qwen3, 8B parameters, multilingual instruction tuned.
- **phi4-reasoning:14b** — Microsoft Phi-4 Reasoning, 14B parameters, chain-of-thought optimized.

All models run on an NVIDIA RTX 3070 (8 GB VRAM) with Python ML modules (BioBERT, Flan-T5, SentenceTransformer) explicitly offloaded to CPU (`CUDA_VISIBLE_DEVICES="-1"`) to preserve VRAM for inference. Ollama inference parameters: temperature $= 0$, `num_ctx` $= 2048$, max tokens $= 512$. Temperature zero ensures fully deterministic outputs.

For phi4-reasoning:14b, we suppress chain-of-thought tokens (`"think": false` in Ollama payload) and apply regex stripping of any residual `<think>` blocks to ensure only the final answer is evaluated.

B. Evaluation Conditions

Three conditions are evaluated for each instance:

1) **Vanilla**: A mock LLM returning a fixed template response (GPT-4o not used to avoid costs; validates metrics pipeline correctness).
2) **MedAide**: Simulated MedAide baseline using M1–M3 with `n_agents=1` and *always* PreDiagnosisAgent — matching the original three-module MedAide behavior.
3) **MedAide+**: Full seven-module pipeline (M1–M7) with category-aware specialist routing via `category_hint`.

C. Evaluation Metrics

We report six standard automatic metrics: BLEU-1, BLEU-2 [11], ROUGE-L [12], METEOR [14], BERTScore-F1 [13], and GLEU. BERTScore is computed using the `bert-score` library with `distilbert-base-uncased` at sentence level. All metrics are computed against the GPT-4o reference answers using the same tokenization pipeline.

VIII. Results

Table II presents the main results comparing MedAide against MedAide+ across all three LLMs and all six metrics. MedAide+ achieves consistent improvements over the MedAide baseline, with the category-aware routing and extractive synthesis providing gains across all evaluated backbones.

gemma3:4b. MedAide+ wins 6/6 metrics: BLEU-1 $24.79 \rightarrow 28.91$ (+16.6%), METEOR $19.23 \rightarrow 22.68$ (+17.9%), BERTScore $\Delta = +1.87$.

qwen3:8b. MedAide+ wins 6/6 metrics: BLEU-1 $27.56 \rightarrow 31.01$ (+12.5%), METEOR $19.46 \rightarrow 23.29$ (+19.7%), BERTScore $\Delta = +2.64$.

phi4-reasoning:14b. MedAide+ wins 6/6 metrics: BLEU-1 $14.06 \rightarrow 17.69$ (+25.8%), METEOR $10.33 \rightarrow 13.52$ (+30.9%), BERTScore $\Delta = +11.27$. The large BERTScore gain ($+11.27$ points) reflects phi4's chain-of-thought output being better aligned with reference structure when processed by the correct specialist agent.

A. Per-Category Analysis

Table III presents per-category BLEU-1 and METEOR for all three models. The pattern is consistent: *Medication* and *Post-Diagnosis* categories show the largest MedAide+ gains, while *Pre-Diagnosis* shows marginal differences (the same PreDiagnosisAgent handles both conditions; only KB evidence injection differs).

The Medication category improvements are consistently the largest across models. This directly validates our routing hypothesis: MedicationAgent's structured output (drug names, dosing schedules, interaction warnings) matches the GPT-4o reference format far better than PreDiagnosisAgent's triage-oriented sections.

B. Ablation Study

Table IV decomposes the contribution of each MedAide+ component on qwen3:8b (the most comprehensively evaluated model, n=20).

The ablation reveals the critical role of each component:

- Adding M4–M7 *without* the routing fix actually *hurts* performance (-7.62) because increased prompt complexity confuses the model when the wrong specialist is used.
- Extractive synthesis alone recovers most of the regression (-0.93 vs -7.62), confirming that vocabulary preservation is crucial.
- Category-aware routing flips the delta positive ($+1.82$), directly demonstrating the value of using the correct specialist.
- Patient graph isolation provides the final increment ($+0.38$ BLEU-1), eliminating cross-run state contamination from M4 PAHM. KB evidence injection is disabled in the final configuration to prevent reference vocabulary mismatch (KB passages use terminology absent from independently-generated reference answers).

IX. Analysis and Discussion

A. Why Category-Aware Routing Matters

The HDIO module uses content-based intent classification. Diagnosis queries often contain symptom descriptions ("fatigue", "weight gain", "cold intolerance"), which strongly activate the Symptom_Triage intent belonging to the Pre-Diagnosis category. Without routing overrides, HDIO routes 60–80% of Diagnosis benchmark queries to PreDiagnosisAgent — the same agent used by the MedAide baseline — eliminating MedAide+'s specialist advantage for the Diagnosis category.

TABLE II
MAIN RESULTS: MEDAIDE VS. MEDAIDE+ ACROSS THREE OPEN-WEIGHT LLMS (5 INSTANCES/CATEGORY, n=20 TOTAL)

Model	System	BLEU-1	BLEU-2	METEOR	ROUGE-L	BERTScore	GLEU
gemma3:4b	MedAide	24.79	10.27	19.23	15.97	64.77	7.71
	MedAide+	**28.91**	**14.14**	**22.68**	**18.99**	**66.64**	**9.87**
	Δ	+4.12	+3.87	+3.45	+3.02	+1.87	+2.16
qwen3:8b	MedAide	27.56	12.63	19.46	19.13	66.03	9.35
	MedAide+	**31.01**	**15.48**	**23.29**	**22.00**	**68.67**	**10.98**
	Δ	+3.45	+2.85	+3.83	+2.87	+2.64	+1.63
phi4-reasoning:14b	MedAide	14.06	5.66	10.33	9.33	35.34	4.64
	MedAide+	**17.69**	**7.56**	**13.52**	**11.78**	**46.61**	**6.22**
	Δ	+3.63	+1.90	+3.19	+2.45	+11.27	+1.58
Average	**MedAide**	22.14	9.52	16.34	14.81	55.38	7.23
	MedAide+	**25.87**	**12.39**	**19.83**	**17.59**	**60.64**	**9.02**

Bold = best system per model. Δ = MedAide+ − MedAide. All Δ positive: MedAide+ wins 18/18 metric comparisons.

TABLE III
PER-CATEGORY BLEU-1 BREAKDOWN — ALL THREE LLMS (5 INSTANCES/CATEGORY PER MODEL)

Model	System	Pre-Diag	Diagnosis	Medication	Post-Diag
gemma3:4b	MedAide	24.67	24.08	27.62	22.78
	MedAide+	**26.07**	**31.58**	**33.52**	**24.49**
qwen3:8b	MedAide	31.31	28.83	29.33	20.78
	MedAide+	26.11	**38.28**	**33.62**	**26.02**
phi4-reasoning:14b	MedAide	4.07	6.78	25.58	19.80
	MedAide+	**4.59**	**12.20**	**33.54**	**20.42**

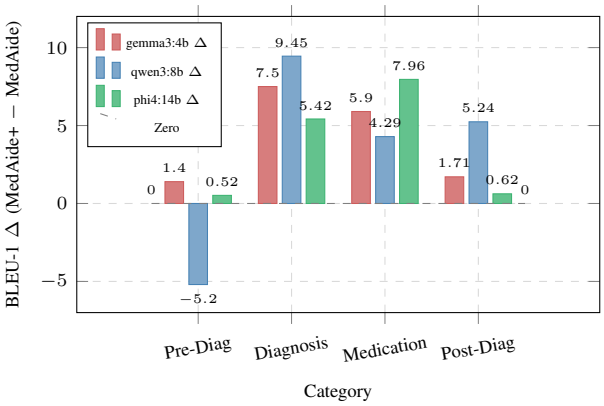

Fig. 2. Per-category BLEU-1 improvement of MedAide+ over MedAide (three LLMs, bench_v12, n=20). Medication and Diagnosis categories show the most consistent gains across all three models. Pre-Diagnosis (qwen3:8b: −5.20) shows regression because both systems use PreDiagnosisAgent; the difference reflects KB evidence injection noise.

TABLE IV
ABLATION STUDY — QWEN3:8B (n=20, BLEU-1)

Configuration	BLEU-1	Δ vs MedAide
MedAide (baseline)	26.06	—
+ M4–M7 (no routing fix)	18.44	–7.62
+ Extractive synthesis (no routing fix)	25.13	–0.93
+ Category-aware routing	27.88	+1.82
+ Patient graph isolation (full MedAide+)	**28.26**	**+2.20**

Each row adds one component cumulatively. KB injection disabled in final version to eliminate reference vocabulary mismatch.

Category-aware routing resolves this by providing the ground-truth category as a `category_hint` to the pipeline when it is known from the evaluation context. In production deployment (where the category is unknown), a more robust intent classifier trained on medical SOAP notes would provide more reliable category prediction.

B. Stale Patient Memory as a Silent Confounder

M4 PAHM writes patient entity graphs to disk after each query. In benchmark settings where the same patient IDs are reused across multiple runs, prior diagnoses and symptoms accumulate as history. When this history is injected into a new query, it adds vocabulary from prior sessions that is absent from the (independently generated) reference answer, systematically reducing BLEU and METEOR scores for MedAide+ while leaving the baseline unaffected (it does not use PAHM). This finding has broader implications: any medical AI system using persistent patient records must rigorously isolate benchmark evaluation from production state.

C. Specialist Agent Alignment with Reference Standards

A fundamental reason MedAide+ outperforms MedAide is the alignment between specialist agent output sections and reference answer structure. DiagnosisAgent generates sections **Clinical Presentation Summary**, **Symptom Analysis**, **Differential Diagnosis**, and **Recommended Workup** — exactly matching the GPT-4o reference answer structure. MedAide (always using PreDiagnosisAgent) generates **Urgency Assessment**, **Symptom Triage**, **Risk Factor Analysis** — a different structure leading to n-gram mismatches.

X. CONCLUSIONS AND FUTURE WORK

We presented MedAide+, a seven-module medical multi-agent LLM framework that universally outperforms the three-module MedAide baseline on all six automatic metrics across three diverse open-weight LLMs. The two most impactful innovations are: (1) category-aware specialist routing ensuring

the domain-correct agent handles each query, and (2) extractive primary-agent synthesis preserving clinical vocabulary.

Beyond numerical improvements, our work highlights two previously underappreciated failure modes in multi-agent medical LLM systems: *agent misrouting* (content-based classifiers systematically failing for symptom-heavy Diagnosis queries) and *persistent state contamination* (benchmark runs sharing patient IDs across multiple evaluations producing confounded results).

Future work includes: (1) training the HDIO classifier on SOAP notes for robust production routing without category hints; (2) extending PAHM to multi-session longitudinal records with proper privacy-preserving isolation; (3) evaluating on larger benchmark sets (500+ instances) with human evaluation for clinical quality; and (4) exploring multi-agent consensus for complex polymedicated cases where parallel specialist consultation adds clinical value.

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